

SECTION 03 54 00

SELF-LEVELING CAST UNDERLAYMENT TOPPING

PART 1 GENERAL

1.1 SECTION INCLUDES

- I. Pumpable and pourable, low-alkali, cement-based, self leveling underlayment system:

1.2 RELATED SECTIONS

- I. Section 03 30 00 - Cast-in-Place Concrete.
- II. Section 06 11 16 - Mechanically Graded Lumber.

1.3 REFERENCES

- I. American Concrete Institute (ACI) 117 - Standard Specifications for Tolerances for Concrete Construction and Materials.
- II. American Concrete Institute (ACI) 318 - Building Code Requirements for Structural Concrete and Commentary.
- III. ASTM International (ASTM):
 1. ASTM C 1077 - Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation.
 2. ASTM C1583 - Standard Test Method for Tensile Strength of Concrete Surfaces and the Bond Strength or Tensile Strength of Concrete Repair and Overlay Materials by Direct Tension (Pull-off Method).
 3. ASTM C1708 - Standard Test Methods for Self leveling Mortars Containing Hydraulic Cements.
 4. ASTM E 329 - Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection.
 5. ASTM F2170 - Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes.
 6. ASTM F3010 - Standard Practice for Two-Component Resin Based Membrane-Forming Moisture Mitigation Systems for Use Under Resilient Floor Coverings
- IV. International Concrete Repair Institute (ICRI):
 1. ICRI Guideline No. 03739 - Field Applications of In-Situ Tensile Pull-Off Tests.

2. ICRI Guideline No. 03732 - Selecting and Specifying Concrete Surface Preparation for Coatings, Sealers, and Polymer Overlays.
- V. Underwriter's Laboratory (UL) 2818 - Green guard - Certification Program for Chemical Emissions for Building Materials, Finishes and Furnishings.

1.4 SYSTEM DESCRIPTION

Single component cementitious floor self leveling underlayment composed It is supplied as a dry powder in pre-weighed bags ready to use on site, once mixed it will form a cohesive self smoothening honey-like flowing mortar that is applied in thicknesses between 2 and 10 mm giving a super flat finish capable of placement over concrete, wood and other types of sound flooring before installing wood, resilient, cork, sports, ceramic, stone, carpet or other flooring systems.

SUBMITTALS

- I. Submit under provisions of Section 01 30 00 - Administrative Requirements.
- II. Product Data: Manufacturer's data sheets on each product to be used, including:
 1. Preparation instructions and recommendations.
 2. Storage and handling requirements and recommendations.
 3. Installation methods.
 4. Contractor's certification of LEED Compliance: Submit Contractor's certification verifying the installation of specified LEED Compliant products.
 5. Product Cut Sheets for all materials that meet the LEED performance criteria. Submit Product Cut Sheets with Contractor or Sub-contractor's stamp, as confirmation that submitted products were installed on Project.
 6. Material Safety Data Sheets for all applicable products.
- III. Submit proof of warranty.
- IV. Submit Health Product Declarations (HPD) for each tile installation material.

1.5 QUALITY ASSURANCE

- I. Manufacturer Qualifications: Minimum 5 year experience manufacturing similar products.
- II. Installer Qualifications: Minimum 2 year experience installing similar products.
- III. Installation of Vetotop CL530 Underlayment's shall be by a qualified applicator using specialized mixing equipment and tools approved by the manufacturer.
- IV. Testing Agency Qualifications: If required, secure an independent agency qualified according to ASTM C 1077 and ASTM E 329 for testing indicated, as documented according to ASTM E 548.
- V. Mock-Up: Provide a mock-up for evaluation of surface preparation techniques and application workmanship.
 1. Place mock-ups of VETOTOP CL530 Underlayment of approximately 10 sq. m.

to demonstrate typical joints, surface finish, bonding, texture, tolerances, and standard of workmanship.

2. Do not proceed with remaining work until workmanship is approved by Architect.
3. Refinish mock-up area as required to produce acceptable work.

- VI. Underlayment is a cement-based product and may exhibit slight variations in colour as a function of job-site conditions. Water-marks similar to veins in stone are a natural result of a pour/spread/smooth installation process.

1.6 PRE-INSTALLATION MEETINGS

- I. Reinstallation Conference: Conduct conference at Project site to comply with requirements in Division 01.

1.7 DELIVERY, STORAGE, AND HANDLING

- I. Deliver materials in their unopened packages and protect from moisture. Protect liquids from freezing and from excessive heat. Store off the floor on dry pallets or equivalent. VETOTOP CL530 is additionally packaged in 25kg bags and shall be used within one year of the date of manufacture.
- II. Use all means necessary to protect the materials before, during and after installation and to protect the installed work and materials of all other trades.

1.8 PROJECT CONDITIONS

- I. Observe the basic rules of concrete placement and finishing. Substrate temperature shall be a minimum of 5 degrees C during the application and the ambient temperature maintained at 10 degrees to 32 degrees C.
- II. Follow hot weather precautions available from manufacturer's Technical Service Department.
- III. Never mix with cement or additives other than manufacturer's approved products.
- IV. During application and curing underlayment shall not be exposed to rapid air movement from mechanical conditioned air. Air flow from HVAC systems in areas of application shall be directed away from the floor during application of underlayment and at least 24 hours after work is completed.
- V. Foot traffic of any kind shall not be allowed in the work area from commencement of substrate preparation until a minimum 24 hours after completion of finishing.

1.9 SEQUENCING

- I. Ensure that products of this Section are supplied to affected trades in time to prevent interruption of construction progress.
- II. Finished floor goods may be installed as soon as one day after application, subject to thickness, drying conditions and type of flooring materials. Always refer to finished floor manufacturer's recommendations regarding installation instructions, restrictions, moisture conditions and compatibility. Always test performance

suitability and compatibility of finished floor systems prior to their application. Sample surfaces shall be installed as a field test so as to be representative of entire surface and tested for intended use.

1.10 WARRANTY

- I. The Contractor warrants the Work of this Section to be in accordance with the Contract Documents and free from faults and defects in materials and workmanship for a period of 5 years. The underlayment manufacturer shall provide a written five years warranty, which covers materials and labor.

PART 2 PRODUCTS

2.1 MANUFACTURERS

- I. Acceptable Manufacturer: VETOTOP CL530, as manufactured by Saudi Vetonit Company Ltd.-Saveto71 St. Second Industrial Estate Riyadh P.O.Box 52235 Riyadh 11563 Saudi Arabia Tel +966 11 2653334, Fax: +966 11 265 0936, E-mail: info.Riyadh@saveto.com; Web: www.saveto.com.

1. Compliance

EN197: 2011
EN16236:2013
EN13813:2002

2. Physical Properties

Compressive Strength @ 28 days (ASTM C349):	> 20 Mpa
Flexural Strength @ 28 days (ASTM C348):	7.5 N/mm ²
Flow @ 10 minutes:	250 mm
VOC	<50 gm/Liter
Dry density (Kg/Liter)	1.80 approx.
Wet density (Kg/Liter)	2.10 approx.
Open time @ 25°C	20 min approx.
Application thickness:	2-10mm

- II. Substitutions: Not permitted.

- III. Requests for substitutions will be considered in accordance with provisions of Section 01 60 00 - Product Requirements.

1. For alternate materials, at least thirty days before bid date submit independent laboratory test results confirming compliance with specified products performance and requirements.

PART 3 EXECUTION

3.1 GENERAL

- I. Concrete Slab Finishing shall be in accordance with ACI 117 (Straight Edge Method).
- II. Provide ventilation to promote curing and reduce humidity.
- III. Do NOT install underlayment over moving cracks in substrate.

3.2 INSPECTION

- I. Verify that installed work of other trades is complete to the point where work of this Section may properly commence.
- II. Verify conformance to Field Conditions specified by this Section and to manufacturer instructions.
- III. Examine the areas and conditions where the Underlayment is to be installed and notify the Architect of conditions detrimental to the proper and timely completion of the work. Verify that substrate surfaces are protected from weather, wind, water and clean, dry, unfrozen, do not contain petroleum by-products, or other compounds detrimental to underlayment material bond to substrate. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.
- IV. Beginning work constitutes acceptance of site conditions and responsibility for defective installation caused by prior observable conditions.

3.3 PREPARATION

Surface Preparation:

1. Clean surface to remove grease, wax, oil or other contamination
2. Larger areas can be divided with wooden battens to control the pumping volume and determine the layer thickness to be applied.
3. Prime the surface with Vetoprime PB433 diluted with potable water (ratio of 1:1.5). Mix thoroughly and spray or brush onto the floor without pooling

1. Cleaning:

- i. Remove dirt, wax, sealers, curing compounds, efflorescence, and other unsuitable surface conditions not part of concrete matrix that may inhibit the underlayment bond to the surface.
- ii. Broom clean and vacuum surfaces to pick up remaining dust and debris.

2. All cracks in the subfloor shall be repaired to minimize telegraphing through the Underlayment.
3. Substrate temperature shall be a minimum of 5 degrees C during the application and the ambient temperature maintained at 10 degrees to 32 degrees C).
4. Concrete slabs shall be tested and measure less than 95% relative humidity (RH) per ASTM F2170 prior to installation of VETOTOP CL530 self-leveling underlayment. If 95% RH or greater is measured contact VETOTOP CL530 technical support. Do not install VETOTOP CL530 products when there is standing water or visible moisture on the slab. It is important to note that concrete slabs may be dry enough to successfully install VETOTOP CL530 products; however, concrete slab may not be dry enough to meet moisture conditions required for finish flooring.
5. Surface bond strength of the substrate shall be no less than 100 PSI (0.7 MPa) per ASTM C1583 or ICRI Guideline No. 03739 for the application of all VETOTOP CL530. Weak or contaminated surfaces shall be mechanically cleaned by shot blasting, scarifying or sanding. Never use chemicals to clean the substrate. If installing VETOTOP CL530 Moisture Vapor Control, per ASTM F3010, concrete slab to receive VETOTOP CL530 Moisture Vapor Control shall have a tensile pull-off strength of 1.4 MPa or greater when tested in accordance with ASTM C1583.
6. Containment: Close and seal floor openings and install dams at edges of floor area to receive treatment as necessary to contain self leveling underlayment while in plastic state.
7. Concrete Floor Flatness and Floor Levelness Benchmarks: Use a digital level device to determine elevations on a 4x4 foot grid, depending on tolerances, to establish and set self-adhering LEVEL PEGS at heights to indicate installation depths and top surface of underlayment application or other approved method.

8. Joint Preparation:

- i. Expansion and Isolation Joints: Honor through underlayment by marking with screws or similar and saw cutting after self-leveling application.
- ii. Static (Non-Moving) Saw Cuts and Control Joints: Fill with joint sealer under provisions of Section 03 30 00 - Cast-in-Place Concrete.
- iii. Other Static (Non-Moving) Joints: Patch and repair using cementitious patching product.
- iv. Dynamic (Active) Cracks: Bring to Architects attention for direction.

9. Priming:

- i. Maintain adequate ventilation during and following primer application to promote faster drying
 1. Insufficient drying time, low temperatures, and high humidity may result insufficient drying, poor film formation, and pinholes in surface.
 2. Do not apply at surface temperatures below 5 degrees C).
 - ii. Primer: Dilute Vetoprime PB433 diluted with potable water (ratio of 1:1.5). Mix thoroughly and spray or brush onto the floor without pooling.
- II. Install underlayment in accordance with manufacturer's instructions, and after installation of moisture vapor mitigation (if needed) and underlayment primer.
 - III. Mixing: Measure components and mechanically mix, as recommended by the manufacturer.
 - IV. Substrate temperature shall be a minimum 40 degrees F (4 degrees C) during application and air temperature maintained between 50 to 90 degrees F (10 to 32 degrees C). Protect areas from direct sunlight and exposure to weather.
 - V. Pump or pour VETOTOP CL530 onto substrate at an thickness ranging from 3-10mm.
 - VI. Immediately following placement lightly smooth the surface and pour lines. When not using level pegs the use of a gauge rake will assist in controlling material depth.
 - VII. Adequate ventilation shall be provided to ensure uniform drying.
 - VIII. Do not expose VETOTOP CL530 to rolling dynamic loads, such as forklifts or scissor lifts, for at least 72 hours after installation.
 - IX. Proper application is the responsibility of the user.
 1. Place underlayment before partition installation if construction schedule allows.
 - X. If a fine, feathered edge is desired, steel trowel the edge after initial set, but before it is completely hard.

3.4 CURING

- I. Once underlayment starts to set, prohibit foot traffic until final set has been reached.
- II. Air cure in accordance with manufacturer's instructions.
- III. Provide continuous ventilation and adequate heat until walkable; provide mechanical ventilation if necessary.
- IV. Remove and replace underlayment areas that evidence lack of bond with substrate,

including areas that emit a "hollow" sound when tapped.

- V. Do not install floor coverings over underlayment until all conditions required by the floor covering manufacturer have been met. Floor covering manufacturer's requirements may vary greatly.

3.5 PROTECTION

- I. Protect against direct sunlight, heat, and wind; prevent rapid drying to avoid shrinkage and cracking during the curing process.
- II. Protect underlayment from traffic as follows:
 - 1. Do not permit foot traffic over underlayment surfaces for 2 to 4 hours, 24 hours for trade traffic.
 - 2. Do not permit trade vehicle traffic for 48-hours.
 - 3. Standing water is to be removed by squeegee or other acceptable method daily.

3.6 FIELD QUALITY CONTROL

- I. Field flow tests shall be performed on site periodically to ensure mix is homogeneous and free from separation.
- II. Provide written reports to Architect.

3.7 PROTECTION

- I. Protect installed products until completion of project.
- II. Touch-up, repair or replace damaged products before Substantial Completion.

END OF SECTION